

8. Functions of Operating System: Memory Management

Computer Science Fundamentals | Module 2: Memory Management | CS_Fundamental-00

Reference to Previous Concepts: Concept 4.2.2 mein humne Kernel ke 5 core functions mein Memory Management dekha tha, aur Concept 3.3 ke numerical example ($a = b + c$) mein RAM allocation samjha tha. Yeh topic **Image 8 (bottom slide)** par aadharit hai.

8.1 Primary Memory / Main Memory (RAM) ki Sanrachna

- **Array of Bytes / Words:** Main memory ek bahut bada array of bytes/words hoti hai jisme har byte/word ka ek unique numerical memory address hota hai.
- **Direct CPU Access:** Fast storage jise CPU system bus ke zariye direct read/write access karta hai.
- **Execution Shart:** Kisi bhi program ko execute hone ke liye uska pehle **Main Memory (RAM)** mein load hona anivarya (mandatory) hai.

Fig 8.1: Main Memory Array Architecture & OS Management Activities

Reference: Image 8 [Primary Memory Array of Bytes, Multiprogramming & Allocation]

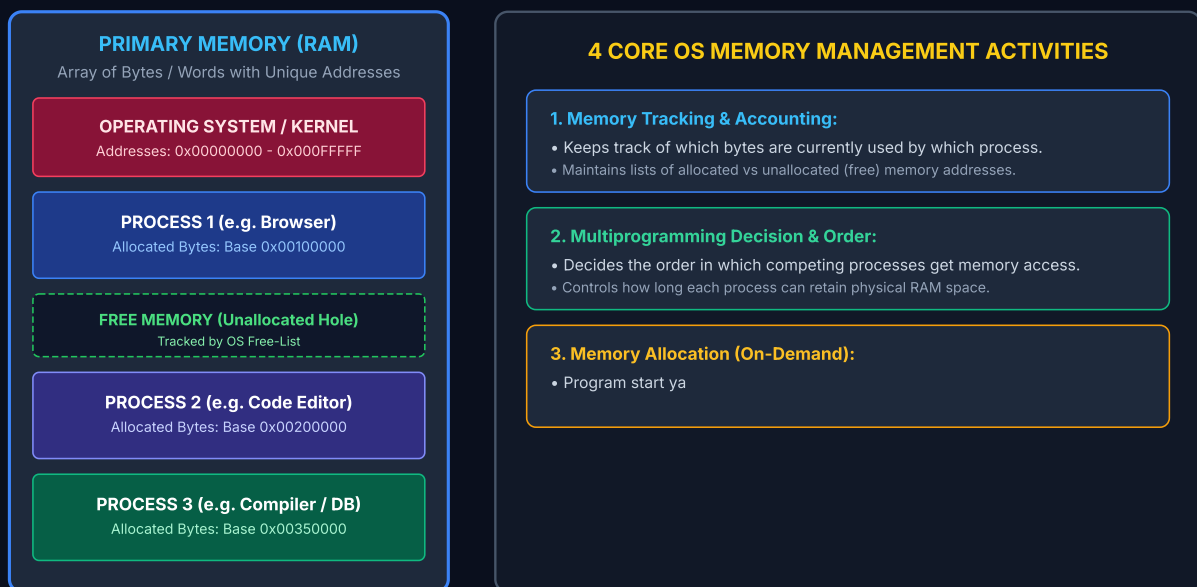


Fig 8.1: Primary Memory Array & 4 Core OS Activities

8.2 OS ki 4 Core Memory Management Activities (Image 8 Reference)

- 1. Memory Tracking & Accounting:** Track karta hai ki kaun se bytes kis process dwara use ho rahe hain, kaun se allocated hain aur kaun se free hain.

2. **Multiprogramming Access Decisions:** Multiprogramming environment mein decide karta hai ki competing processes ko kis order mein aur kitni der tak memory access milega.
3. **Dynamic Memory Allocation:** Process request par RAM bytes allot karta hai aur MMU base/limit registers set karta hai.
4. **Memory Deallocation & Reclamation:** Process terminate hone par memory reclaim karta hai taaki Memory Leaks na hon.

8.3 Numerical / Computational Example & Step-by-Step Thought Process: Memory Addressing & Allocation

Scenario: Total RAM = 1024 KB. OS Reserved = 0-199 KB (200 KB). Free = 200-1023 KB (824 KB).

1. **Process P1 Request (300 KB):** OS P1 ko 200 se 499 KB allot karta hai. Base Register = 200 KB, Limit Register = 300 KB. Free memory bachi = 524 KB (500-1023 KB).
2. **Process P2 Request (250 KB):** OS P2 ko 500 se 749 KB allot karta hai. Base Register = 500 KB, Limit Register = 250 KB. Free memory bachi = 274 KB.
3. **Hardware Protection Verification:** Agar P2 address 450 KB access karne ki koshish kare, toh MMU hardware check karega: $450 < \text{Base}(500)$ → **Illegal Access!** Trap trigger karke process kill ho jayega (Memory Protection!).
4. **Deallocation:** P1 terminate hone par 300 KB block free-list mein return hota hai aur agle processes ke liye ready ho jata hai.